

THE EXPONENT SPECTRUM OF PERFECT-POWER VALUES OF POLYNOMIALS, WITH COMPLETE RUNGE ENUMERATION AND A MACHINE-CHECKED KERNEL

PERFECTPOWER PROJECT

ABSTRACT. For $F \in \mathbb{Z}[x]$ and $d \geq 2$ let $A(N)$ be the number of $1 \leq n \leq N$ for which $F(n)$ is a perfect d -th power. We classify (F, d) into four types, decidable from the root multiplicities of F . In the four types, $A(N) = N$; $A(N) = \kappa N^{1/t} + O(1)$ with $t \mid d$; $A(N) = \kappa \log N + O(1)$; or $A(N) = O(1)$. The constants κ are explicit. The first three types are handled by elementary arguments, and the last by LeVeque's theorem. Consequently the growth exponent $\limsup \log(1 + A(N))/\log N$ lies in $\{0, 1\} \cup \{1/t : t \mid d, t > 1\}$, and $A(N) = O(N^{1/p})$ for p the least prime factor of d whenever F is not a d -th power in $\mathbb{Z}[x]$. In the Runge case ($d \mid \deg F$ and $\text{lc } F$ a d -th power) we give a complete enumeration of the hit set by exact root isolation. We describe the dependence on a shift $F = S + k$, and give Dirichlet-series and heat-kernel asymptotics for every type. Several results, among them the rigid dichotomy, the rigid 0–1 law, the monomial count and the Pell example $2n^2 + 1$, are verified in Lean 4 with Mathlib.

1. INTRODUCTION

Let $F \in \mathbb{Z}[x]$ and $d \geq 2$. Call $n \geq 1$ a *hit* if $F(n) = m^d$ for some $m \in \mathbb{Z}$, and let $A(N) = \#\{1 \leq n \leq N : n \text{ is a hit}\}$. The density $\lim A(N)/N$ always exists: it is 1 if $F = G^d$ with $G \in \mathbb{Z}[x]$ and 0 otherwise. This follows, for instance, from Boshernitzan's equidistribution theorem for Hardy fields together with a separate treatment of the “rigid” case, and this was the starting point of the project.

Density zero is a weak statement. The hit sets of $4n + 1$, of $2n^2 + 1$ and of $n^3 + 1$ (with $d = 2$) have respectively about $\sqrt{N}$ elements, about $0.567 \log N$ elements, and finitely many elements. The purpose of this paper is to show that these three behaviours, together with density one, are the only possibilities, and to compute everything explicitly.

Setup. Write $F = c \prod_i (x - \alpha_i)^{r_i}$ over $\overline{\mathbb{Q}}$ with distinct α_i and $c = \text{lc } F$. Put $t_i = d / \gcd(d, r_i)$. The multiset $\{t_i\}$ is the t -profile of (F, d) . It is determined by Yun's squarefree decomposition $F = c \prod_j S_j^{j_j}$: every root of S_j has $t = d / \gcd(d, j)$. No factorisation is needed.

Theorem 1.1 (Atlas). *Let $F \in \mathbb{Z}[x]$ be nonconstant and $d \geq 2$. Exactly one of the following holds.*

- (i) **Power type:** all $t_i = 1$. Then $A(N) = N$ if c is an integer d -th power, and otherwise the hits are exactly the positive integer roots of F .
- (ii) **Radical type:** the t -profile is $\{t, 1, \dots, 1\}$ with $t \geq 2$. Then $A(N) = \kappa N^{1/t} + O(1)$ with $\kappa \geq 0$ given by (1).
- (iii) **Pell type:** the t -profile is $\{2, 2, 1, \dots, 1\}$. Then $A(N) = \kappa \log N + O(1)$ with $\kappa \geq 0$ given by (2).
- (iv) **Finite type:** all remaining cases. Then the hit set is finite.

Corollary 1.2 (Exponent spectrum). *Let $\alpha(F, d) = \limsup_N \log(1 + A(N))/\log N$. Then $\alpha(F, d) \in \{0, 1\} \cup \{1/t : t \mid d, t \geq 2\}$, and each value occurs. If F is not the d -th power*

of a polynomial in $\mathbb{Z}[x]$, then $A(N) = O(N^{1/p})$, where p is the least prime factor of d . In particular $A(N) = O(\sqrt{N})$ always, and $A(N) = O(N^{1/3})$ for odd d . These bounds are attained.

Proof. In type (ii) the integer $t = d/\gcd(d, r)$ divides d and exceeds 1, so $t \geq p$. Types (iii) and (iv) have exponent 0. The monomial x^r with $\gcd(r, d) = d/t$ has $A(N) = \lfloor N^{1/t} \rfloor$ by Theorem 3.1. $\square$

Parts (i)–(iii) of Theorem 1.1 are proved in §§2–4. Part (iv) is LeVeque’s theorem [4]. We therefore obtain a second proof of the density 0–1 law, independent of equidistribution. It depends on Siegel’s theorem through LeVeque, but it is quantitative. Section 5 makes type (iv) effective in the Runge case. Section 6 treats shifts, §7 treats transforms, and §8 describes the Lean formalisation and the computations.

2. THE POWER TYPE

Theorem 2.1. *Suppose all root multiplicities of F are divisible by d , so that $F = cG^d$ with $G \in \mathbb{Q}[x]$ monic. If $c = b^d$ with $b \in \mathbb{Z}$, then $bG \in \mathbb{Z}[x]$ and every n is a hit. Otherwise $n \geq 1$ is a hit if and only if $F(n) = 0$.*

Proof. If $c = b^d$, then $(bG)^d = F \in \mathbb{Z}[x]$. Since $\mathbb{Z}[x]$ is integrally closed in $\mathbb{Q}[x]$ (Gauss’s lemma), $bG \in \mathbb{Z}[x]$. Now suppose c is not an integer d -th power. If $G(n) \neq 0$ and $F(n) = m^d$, then $c = (m/G(n))^d$. A rational number whose d -th power is an integer is an integer, so c would be an integer d -th power, a contradiction. Finally, $F(n) = 0 = 0^d$ whenever $G(n) = 0$. $\square$

Remark 2.2 (Grunwald–Wang). Take $F = 16x^8$ and $d = 8$. Since 16 is an 8-th power modulo every odd prime, $F(n)$ is an 8-th power residue modulo every prime for every n . Yet F has no hits. No sieve that uses residues modulo primes alone can prove density zero in general. Here the obstruction is modulo 32.

3. THE RADICAL TYPE

Galois-conjugate roots of F have equal multiplicities. A unique root with $t \geq 2$ is therefore rational, say $\alpha = u/v$ with $v \geq 1$ and $\gcd(u, v) = 1$. We can write

$$F = c(x - \alpha)^r G(x)^d, \quad G \in \mathbb{Q}[x] \text{ monic}, \quad t = d/\gcd(r, d) \geq 2.$$

Let $K = \lceil r/d \rceil$ and $c_1 = cv^{dK-r} \in \mathbb{Z}$. Let $g = \gcd(r, d)$ and $r' = r/g$, so $\gcd(r', t) = 1$. If $g \mid v_p(c_1)$ for all primes p , let $\tau_p \in [0, t)$ be defined by $\tau_p \equiv -(v_p(c_1)/g)r'^{-1} \pmod{t}$, and put $z_0 = \prod_p p^{\tau_p}$. A sign $\sigma \in \{\pm 1\}$ is *admissible* if d is odd, or if d is even and $c_1\sigma^r > 0$.

Theorem 3.1. *In the radical type, $n \geq 1$ is a hit if and only if either $F(n) = 0$, or $g \mid v_p(c_1)$ for every p and $vn - u = \sigma z_0 w^t$ for an admissible sign σ and an integer $w \geq 1$. Let $\varrho = \#\{w \pmod{v} : z_0 w^t + u \equiv 0 \pmod{v}\}$. If $+1$ is admissible and the divisibility condition holds, then*

$$(1) \quad A(N) = \kappa N^{1/t} + O(1), \quad \kappa = \frac{\varrho}{v} \left(\frac{v}{z_0} \right)^{1/t}.$$

Otherwise $A(N) = O(1)$.

Proof. Let $F(n) \neq 0$. We claim that n is a hit if and only if $c(n - \alpha)^r \in \mathbb{Q}^{\times d}$. If $F(n) = m^d$, divide by $G(n)^d$. Conversely, if $c(n - \alpha)^r = \xi^d$, then $F(n) = (\xi G(n))^d$ is an integer and a rational d -th power, hence an integer d -th power.

With $z = vn - u \neq 0$, multiplying by the d -th power v^{dK} shows that the condition is equivalent to $c_1 z^r$ being the d -th power of an integer. An integer $X \neq 0$ is a d -th power if and only if $d \mid v_p(X)$ for every p , together with $X > 0$ when d is even. For $X = c_1 z^r$ the valuation condition

is $v_p(c_1) + rv_p(z) \equiv 0 \pmod{d}$. This is solvable exactly when $g \mid v_p(c_1)$, and then it is equivalent to $v_p(z) \equiv \tau_p \pmod{t}$. The positive integers with these valuations are exactly $z_0 w^t$ with $w \geq 1$.

For the count, the sign $\sigma = +1$ contributes $n = (z_0 w^t + u)/v$ with w in one of ϱ residue classes modulo v and $z_0 w^t \leq vN - u$. That gives $\varrho v^{-1}((vN - u)/z_0)^{1/t} + O(1)$ hits. The sign $\sigma = -1$ forces $n < \alpha$, and there are finitely many zeros of F . $\square$

Example 3.2. • $4n + 1$ with $d = 2$: $\kappa = 1$.

- $2n - 5$ with $d = 2$: $\kappa = 2^{-1/2}$.
- $3n + 5$ with $d = 3$: $\kappa \approx 0.4807$.
- $12n^2$ with $d = 3$: the hits are $n = 12w^3$, so $\kappa = 12^{-1/3}$.
- $3n + 2$ with $d = 6$ and $16n^3$ with $d = 6$: no hits.

4. THE PELL TYPE

Here $d = 2e$, and the two special roots have multiplicities that are odd multiples of e . They are either both rational or conjugate. Their monic product $W \in \mathbb{Q}[x]$ has distinct roots, and $F = cW^e H^d$ with $H \in \mathbb{Q}[x]$ monic.

Theorem 4.1. *Let $\Gamma = \{\gamma \in \mathbb{Q} : \gamma^e = c\}$, so that $|\Gamma| \leq 2$. For $\gamma = \gamma_1/\gamma_2 \in \Gamma$ in lowest terms, let L be the common denominator of W and put $P_\gamma = \gamma_1 \gamma_2 L \cdot LW \in \mathbb{Z}[x]$. Then $n \geq 1$ is a hit if and only if $F(n) = 0$ or $P_\gamma(n)$ is a perfect square for some $\gamma \in \Gamma$.*

Proof. For $F(n) \neq 0$, n is a hit if and only if $cW(n)^e = y^{2e}$ with $y \in \mathbb{Q}^\times$. In that case $\gamma = y^2/W(n)$ satisfies $\gamma^e = c$. Conversely, if $\gamma W(n) = y^2$ with $\gamma^e = c$, then $cW(n)^e = y^{2e}$. Multiplying by $(\gamma_2 L)^2$ turns “ $\gamma W(n)$ is a rational square” into “ $P_\gamma(n)$ is a perfect square”. $\square$

Lemma 4.2. *Let $P = An^2 + Bn + C \in \mathbb{Z}[x]$ with $A \neq 0$ and $\Delta = B^2 - 4AC \neq 0$.*

- (a) *If $A < 0$ or A is a square, then $P(n)$ is a square for only finitely many n .*
- (b) *If $A > 0$ is not a square, let $\varepsilon = x_1 + y_1 \sqrt{4A}$ be the fundamental solution of $x^2 - 4Ay^2 = 1$. For each ε -orbit $\mathcal{O}$ of solutions $\eta = X + Y\sqrt{4A} > 0$ of $X^2 - 4AY^2 = \Delta$, let $\pi_{\mathcal{O}}$ be the period of $(X, Y) \pmod{2A}$ along the orbit, and let $g_{\mathcal{O}}$ be the number of steps in one period with $X \equiv B \pmod{2A}$. Then*

$$(2) \quad \#\{n \leq N : P(n) = \square\} = \kappa \log N + O(1), \quad \kappa = \frac{1}{\log \varepsilon} \sum_{\mathcal{O}} \frac{g_{\mathcal{O}}}{\pi_{\mathcal{O}}}.$$

Proof. We have $4AP(n) = X^2 - \Delta$ with $X = 2An + B$. So $P(n) = m^2$ if and only if $X^2 - 4Am^2 = \Delta$, and $n \mapsto X$ is a bijection onto the class $B \pmod{2A}$.

In (a), $A < 0$ bounds X . If $A = s^2$, the factorisation $(X - 2sm)(X + 2sm) = \Delta$ leaves at most $2\tau(|\Delta|)$ possibilities.

In (b) there are finitely many orbits (Nagell). A large hit corresponds to exactly one large $\eta = \eta_0 \varepsilon^k$ in a positive orbit with $X \equiv B \pmod{2A}$. The step $(X, Y) \mapsto (x_1 X + 4Ay_1 Y, y_1 X + x_1 Y)$ has determinant 1, so it is invertible modulo $2A$ and the residues are purely periodic. Finally $\log X_k = k \log \varepsilon + O(1)$. $\square$

Example 4.3. For $2n^2 + 1$ we get $\kappa = 1/\log(3 + 2\sqrt{2}) = 0.56729\dots$. The exact counts at $N = 10^6, 10^{12}, 10^{24}, 10^{48}$ are 8, 16, 31, 63. The square triangular numbers $n(n+1)/2$, reduced to $4n^2 + 4n$ by Remark 4.4, have the same κ . For the rigid polynomials $n^2 + n$ and $n^2 + 1$, A is a square and there are no hits.

Proof of Theorem 1.1. The t -profiles in (i)–(iv) are mutually exclusive and exhaust all possibilities. Parts (i)–(iii) are Theorems 2.1, 3.1 and 4.1 together with Lemma 4.2. Part (iv) is LeVeque’s theorem [4]: the equation $y^m = f(x)$ has only finitely many integral solutions unless the t -profile is $\{t, 1, \dots, 1\}$ or $\{2, 2, 1, \dots, 1\}$. $\square$

Remark 4.4 (Integer-valued polynomials). If $F \in \mathbb{Q}[x]$ is integer-valued with coefficient denominator L , then F and $L^d F \in \mathbb{Z}[x]$ have the same hits. This brings binomial coefficients into scope. For example, $\binom{n}{3}$ is a square for $n \leq 10^5$ only when $n \in \{1, 2, 3, 4, 50\}$.

5. COMPLETE ENUMERATION IN THE RUNGE CASE

Suppose $\deg F = dq$ and $\text{lc } F = b^d$ with $b \in \mathbb{Z}$ (the *rigid* case), and that F is not a d -th power. Matching the top $q + 1$ coefficients gives a truncated root $Q = P/D$ with $P \in \mathbb{Z}[x]$, $D \geq 1$ and $\text{lc } Q = b$, such that $R = F - Q^d \neq 0$ has degree $r < (d - 1)q$. Qualitatively this shows that the hit set is finite. The Lean theorem `rigid_dichotomy` checks this.

Theorem 5.1. *For $x_0 \geq 1$ put*

$$a(x_0) = |b| - \sum_{i < q} |Q_i| x_0^{i-q}, \quad C(x_0) = \sum_{i \leq r} |R_i| x_0^{i-r}, \quad T(x_0) = \left\lfloor \frac{D C(x_0) x_0^{r-(d-1)q}}{a(x_0)^{d-1}} \right\rfloor.$$

Suppose $a(x_0) > 0$ and, when d is odd, $C(x_0) x_0^r < a(x_0)^d x_0^{dq}$. Then every hit $n \geq x_0$ is a root of one of the nonzero integer polynomials

$$G_t = D^d F - (P + t)^d, \quad |t| \leq T(x_0).$$

In particular the number of hits is at most $x_0 - 1 + (2T(x_0) + 1)(d - 1)q$.

Proof. For $n \geq x_0$ we have $|Q(n)| \geq a(x_0) n^q > 0$ and $|R(n)| \leq C(x_0) n^r$. Let $F(n) = m^d$ and pick $y = \pm m$ with $y^d = F(n)$ and $yQ(n) \geq 0$. For odd d the sign condition forces $y = m$ to work, because $|R(n)| < |Q(n)|^d$ gives $F(n)$ the sign of $Q(n)$. Write $y = sQ(n)$ with $s \geq 0$. Then $|s - 1| \leq |s^d - 1| = |R(n)|/|Q(n)|^d$, so

$$|Dy - P(n)| \leq D|R(n)|/|Q(n)|^{d-1} \leq D C(x_0) x_0^{r-(d-1)q} / a(x_0)^{d-1}.$$

Hence $t = Dy - P(n) \in \mathbb{Z}$ satisfies $|t| \leq T(x_0)$ and $G_t(n) = 0$. Finally, $G_0 = D^d R \neq 0$, and for $t \neq 0$ the polynomial G_t has degree $(d - 1)q$ with leading coefficient $-dt (Db)^{d-1}$. $\square$

Since $T(x_0) \rightarrow 0$, the following algorithm is complete. Scan up to the first admissible x_0 . Then walk dyadic blocks $[x_0, 2x_0)$, and in each block either scan it or isolate the integer roots of all G_t with $|t| \leq T(x_0)$ by Sturm sequences, whichever is cheaper. Stop when $T(x_0) = 0$, where only G_0 remains. If F is rigid for some divisor $d' \geq 2$ of d , the same procedure applies, because d -th powers are d' -th powers. By Capelli's theorem this covers the leading-term form of Runge's criterion, up to a trivially negative case.

Example 5.2. $1 + n + n^2 + n^3 + n^4$ is a square for $n \geq 1$ only at $n = 3$ (Ljunggren). For the following pairs, the product of k consecutive positive integers is never a d -th power; each is a fixed instance of the Erdős-Selfridge theorem, and each takes at most 0.03 s:

k	d
4	2, 4
6	2, 3, 6
8	2, 4, 8
10	2, 5
12	2, 3, 4, 6

The earlier cutoff-only method for these pairs needed cutoffs as large as $3.9 \cdot 10^{10}$.

6. SHIFTS

Corollary 6.1. *Let $S \in \mathbb{Z}[x]$ with $\deg S = M \geq 1$, and let $\mathcal{K}(S)$ be the set of integer critical values $-S(\xi)$, where $S'(\xi) = 0$. It has at most $M - 1$ elements, and it is the set of integer roots of $\text{Res}_x(S + k, S')$. For $k \notin \mathcal{K}(S)$, the polynomial $S + k$ is squarefree and is:*

- of radical type if $M = 1$;
- of Pell type if $M = 2$ and $d = 2$;
- of finite type otherwise.

In particular, if $M \geq 3$ then $S(n) + k$ has finitely many hits for all but at most $M - 1$ values of k , and at most one k gives density one.

For example, $n^3 + k$ with $d = 2$ has finitely many hits for every $k \neq 0$; these are the Mordell curves. For $n^3 - 3n + k$ with $d = 2$ the exceptional shifts are $k = \pm 2$, and $n^3 - 3n + 2 = (n - 1)^2(n + 2)$ has $\kappa = 1$.

7. TRANSFORMS

Let X be the hit indicator, $Z_X(s) = \sum X(n)n^{-s}$ and $K_X(\tau) = \sum X(n)e^{-\tau n}$.

Theorem 7.1. *If $A(x) = \kappa x^\beta + O(1)$ with $0 < \beta \leq 1$, then $Z_X(s) - \kappa s/(s - \beta)$ is holomorphic on $\Re s > 0$, and $K_X(\tau) = \kappa \Gamma(1 + \beta) \tau^{-\beta} + O(1)$. If $A(x) = \kappa \log x + O(1)$, then $Z_X(s) - \kappa/s$ is holomorphic on $\Re s > 0$, and $K_X(\tau) = \kappa \log(1/\tau) + O(1)$.*

Proof. Use Abel summation, $Z_X(s) = s \int_1^\infty A(x)x^{-s-1} dx$ and $K_X(\tau) = \tau \int_0^\infty A(x)e^{-\tau x} dx$. The main terms integrate explicitly, and a bounded error contributes a function that is holomorphic on $\Re s > 0$, respectively bounded. $\square$

By Theorem 3.1 and Lemma 4.2 these hypotheses hold exactly in the radical and Pell types. When $\alpha = 0$ in the radical type, one gets $Z_X(s) = z_0^{-s} \zeta(ts)$ plus a Dirichlet polynomial.

8. FORMAL VERIFICATION AND COMPUTATION

The Lean 4 library in the repository compiles against Mathlib (tag v4.20.0). An audit script checks that 56 theorems depend only on `propext`, `Classical.choice` and `Quot.sound`. Among them:

- `rigid_dichotomy`: in the rigid case, either $F = G^d$ over $\mathbb{Z}$ or the hit set is finite.
- `rigid_zero_one`: in the rigid case the density exists and is 0 or 1.
- `twisted_finite`: if $D^d F = cH^d$ with c not a d -th power, every hit is a root of H .
- `monomial_count`: for $F = x^r$, $A(N) = \#\{w \leq N : w^t \leq N\}$.
- `ljunggren_hitSet` and `consecutive_four_hitSet`: two complete hit lists.
- `pell_hitSet_infinite` and `pell_hasDensity_zero`: $2n^2 + 1$ has infinitely many square values and density zero.
- `runge_finite` (Theorem 5.1 in integer form) and `power_type_finite` (Theorem 2.1 over $\mathbb{Q}[x]$).
- 17 machine-generated theorems of the form “the hit set of F is exactly H ”. A Python search finds the Runge data (x_0, T) and Taylor-shifted polynomials, and Lean checks every identity by `ring`, every sign by `positivity`, and every small case by `norm_num`. The instances include products of 4, 6, 8, 10, 12 consecutive integers.

The exact Python implementation (standard library only) classifies (F, d) , computes κ , enumerates hits structurally or by Theorem 5.1, and cross-checks every structural count against the defining scan. The receipts record 43 families and are reproducible bit-for-bit.

What remains.

- Effective enumeration in the finite type outside the Runge case. This requires elliptic-logarithm or Baker-type methods; an example is $n^3 + n + 4 = m^2$, which has the isolated hit $n = 4128$.
- Formalisation of Theorems 3.1, 4.1 and 5.1 in general.
- Analogues for non-polynomial sequences.

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